IMF Business School

Master's Degree in Data Science and Business Intelligence

Remote Learning

Introduction to topological data analysis applied to machine learning

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Chapter 1

Introduction

We live in an era where Data Science and Artificial Intelligence have exponentially accelerated our ability to process information and automate decisions. However, the capacity to process larger datasets faster does not guarantee a deeper understanding of the underlying information. As data analysis scales, there is a risk of focusing purely on making existing processes faster while overlooking the need to investigate alternative analytical approaches. To continue advancing and solving complex societal challenges, Data Science must constantly evolve, embracing frameworks capable of revealing geometric properties and relationships that conventional statistical approaches might miss.

One approach that offers a profoundly different perspective is Topological Data Analysis (TDA). By applying concepts from algebraic topology, TDA investigates the fundamental shape of data, analysing properties such as connectivity, clusters, and multi-dimensional holes. Through techniques like persistent homology, TDA tracks how these geometric features evolve and persist across different scales. This perspective allows us to investigate the organisation of the data as a whole, capturing robust structural patterns that traditional tabular analysis often fails to identify.

The potential application of these ideas is particularly promising for complex and heterogeneous challenges, such as financial fraud detection. In highly imbalanced transaction networks, where fraudulent behaviour accounts for a minute fraction of the data, topological summaries can provide crucial spatial context. Identifying coordinated anomalous behaviours, such as sophisticated fraud rings, requires moving beyond individual variables to understand the global structure of the dataset.

The integration of TDA into machine learning workflows is an active and growing frontier in research. Pioneers in applied topology, such as Gunnar Carlsson [1], have long demonstrated

that structural properties can unlock deep insights in high-dimensional spaces. In the financial sector, topological methods are increasingly being explored for Anti-Money Laundering (AML) and risk management to detect cyclical money flows or synthetic identities [2]. Furthermore, extracting topological metrics, such as distances to persistence loops, and injecting them into ensemble models like XGBoost has proven effective in contemporary studies [3].

These ongoing advancements strongly motivate the need to shed more light on the practical capabilities of TDA in modern Data Science. On a personal level, this motivation is deeply rooted in my academic journey, as I already explored the mathematical foundations of Topological Data Analysis extensively in my Bachelor's thesis [4]. This Master's project represents the natural evolution of that research, transitioning from theoretical mathematics to practical, large-scale machine learning applications. Furthermore, having worked professionally with Amazon Web Services (AWS) for the past two years, I have developed a strong foundation in cloud computing paradigms. Undertaking a project of this magnitude presented a clear opportunity to bridge my academic interests in topology with the scalable cloud engineering skills I have acquired in the industry.

Therefore, the main objective of this Master's thesis is to rigorously investigate the usefulness of TDA in fraud detection, serving as a practical capstone that integrates the analytical methodologies and Data Science best practices acquired during the program. Focusing exclusively on the feature engineering phase, this study evaluates whether appending persistent homology features to the public IEEE-CIS Fraud Detection dataset improves predictive performance over a classical tabular baseline. By executing this end-to-end pipeline entirely within a Google Cloud Platform (GCP) architecture, the thesis builds upon solid theoretical foundations to provide empirical evidence on both the predictive value of TDA and its operational viability in scalable, real-world cloud environments.

The remainder of this thesis is organised as follows:

- **Chapter 2** details the Google Cloud Platform (GCP) architecture and the specific cloud services utilized to build a scalable, end-to-end processing pipeline.
- **Chapter 3** presents a comprehensive Exploratory Data Analysis (EDA) of the raw transactional dataset to understand its underlying distributions and severe class imbalance.
- **Chapter 4** covers the core data processing and feature engineering, detailing the creation of both classical features and the extraction of novel topological variables.
- **Chapter 5** focuses on the training, evaluation, and empirical comparison between the baseline and the TDA-enriched machine learning models.
- **Chapter 6** highlights the main conclusions derived from the study, discusses its limitations, and proposes directions for future research.

Chapter 2

Google Cloud Platform Architecture

To support scalable and reproducible processing, this study leverages **Google Cloud Platform (GCP)**, adopting the core cloud principle of **decoupling compute from storage**. This architectural choice separates data persistence from execution engines, ensuring resources are allocated efficiently. By isolating storage from compute, the pipeline avoids the financial overhead of maintaining continuously running clusters. Processing power scales up dynamically only when executing workloads and scales down to zero immediately after, optimizing both cost-effectiveness and flexibility.

Cloud Persistence Layer: Google Cloud Storage

Google Cloud Storage (GCS) acts as the centralized data lake across the pipeline, providing highly durable, low-cost object storage. Storing data in compressed Apache Parquet format minimizes network I/O bottlenecks and enables seamless, concurrent access across multiple serverless execution engines via standard `gs://` URIs. The storage is structured into a medallion-style architecture:

1. **Raw Tier (Bronze):** Stores immutable CSV files (`transaction.csv` and `identity.csv`).

2. **Processed Tier (Silver/Gold):** Contains intermediate and final Parquet datasets from ETL and TDA extraction.

3. **Artifact Tier:** Houses serialized model binaries and prediction output logs.

Exploratory Data Analysis and Validation via BigQuery

To perform rigorous Exploratory Data Analysis (EDA) and validate data consistency across the ETL pipeline stages, **Google Cloud BigQuery** was utilized as the primary analytical engine. Following data ingestion, raw CSV files stored in Google Cloud Storage were loaded directly into initial BigQuery tables. This serverless data warehouse eliminates the need for database administration or index tuning, enabling lightning-fast SQL queries over millions of rows through its highly parallelized, columnar architecture.

SQL-based EDA was conducted to quantify the severe class imbalance, profile missing values and feature sparsity across columns (revealing that missingness itself is a critical predictive signal, leading to the strategic decision to retain all sparse features), and inspect distributions of temporal variables and transaction amounts to identify skews and anomalies. Furthermore, to ensure strict data integrity and prevent silent failures between the distributed PySpark ETL stages, an incremental validation strategy was implemented in BigQuery. Dedicated SQL view tables were created for each distinct step of the ETL run, such as validating schema integrity post-join, monitoring the robust imputation of missing values, and verifying feature transformations, allowing intermediate outputs to be systematically audited before feeding data into the modeling phase.

Data Engineering via Dataproc Serverless

To perform distributed ETL operations on the IEEE-CIS dataset, **Google Cloud Dataproc Serverless** was utilized as the execution engine for Apache Spark (PySpark). Embracing a fully serverless paradigm, Dataproc completely abstracts underlying infrastructure management. It dynamically allocates the exact compute nodes required to join raw inputs via a `LEFT JOIN`, clean missing values, and serialize datasets into high-performance Apache Parquet files, automatically deprovisioning resources upon job completion. This pay-per-execution model eliminates the operational complexity of tuning persistent Hadoop/Spark clusters. To ensure clean code modularity and reproducibility, the underlying PySpark execution scripts (`.py`) were stored directly in Google Cloud Storage, allowing the Dataproc Serverless jobs to reference and execute them seamlessly from the centralized object repository.

Modeling and Execution Pipeline

Model development and batch evaluation were orchestrated using **Vertex AI** to ensure robust execution and a streamlined MLOps lifecycle. Transitioning from data engineering to machine

learning requires isolated, reproducible environments. Specifically, Vertex AI Custom Training was utilized to train the machine learning models by configuring explicit infrastructure parameters, such as allocating a `n1-standard-8` machine type for the primary worker pool, utilizing custom prebuilt container images, and pointing the job configuration to the specific training script (`.py`) stored in Google Cloud Storage alongside the input datasets. This managed approach guarantees that training environments are completely reproducible and decoupled from the data extraction processes. Finally, **Vertex AI Batch Prediction** evaluated the trained models against the out-of-time test partitions in bulk, generating comprehensive fraud probability scores efficiently without the need to deploy and manage persistent real-time API endpoints.

Chapter 3

Explorative Data Analysis

Credit card fraud detection is a critical challenge in digital payments. Because fraudulent transactions are rare and often share similar characteristics with legitimate ones, accurately identifying suspicious activity requires finding complex patterns while avoiding high rates of false positives. This study explores whether Data Science and Topological Data Analysis (TDA) can provide complementary structural insights into transaction data to enhance fraud detection systems.

The analysis relies on the IEEE-CIS Fraud Detection dataset from Kaggle, a binary classification problem where the target variable, `isFraud`, indicates whether a transaction is fraudulent. The dataset comprises two main complementary files linked by a unique identifier, `TransactionID`: the transaction table (containing economic, temporal, and categorical attributes) and the identity table (containing device and network features).

The following sections describe the preliminary Exploratory Data Analysis (EDA) conducted via SQL queries in Google BigQuery to inspect data structures, variable distributions, and missingness before feature engineering.

To ensure full reproducibility, all underlying assets are publicly available in the project's GitHub repository. The repository is organized into dedicated `identity` and `transaction` folders, which contain the `.sql` queries, their execution results in `.json` format, and the auxiliary Python scripts (`.py`) used to process the data and generate the corresponding `.png` visualizations.

3.1 Identity

The initial phase of the Exploratory Data Analysis focused on the `identity_raw` table, which complements transaction data with digital identity, device, and network attributes (such as `DeviceType`, `DeviceInfo`, and `id_12` to `id_38`). Identity records are available for only a subset of transactions, making missingness a key structural trait of this table.

3.1.1 Dataset Structure and Data Types

The `identity_raw` dataset comprises 144,233 records and 41 variables. The data types are distributed as follows:

- **FLOAT64 (68.29%)**: The clear majority, representing numerical attributes.
- **STRING (19.51%)**: Representing categorical data.
- **BOOL (9.76%)**: Representing binary indicators.
- **INT64 (2.44%)**: Exclusively representing the `TransactionID`.

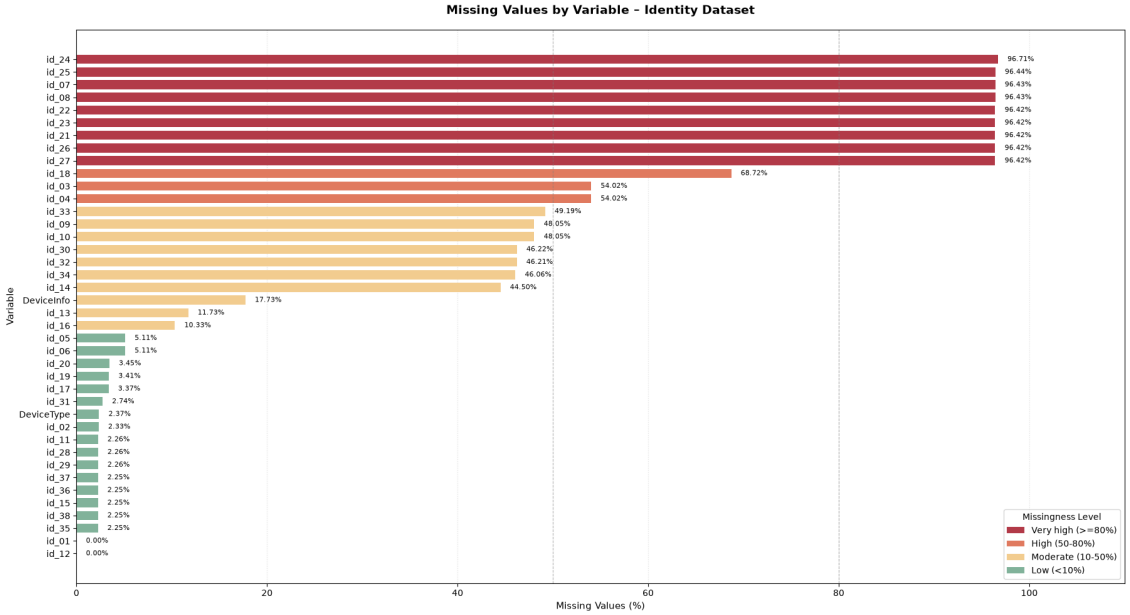
The `TransactionID` variable demonstrates perfect integrity; it contains zero missing values and no duplicate records. Consequently, it serves as a robust primary key for subsequent joins with the transactional data.

3.1.2 Missing Data Assessment

A defining characteristic of the identity dataset is the highly unequal distribution of missing values, which poses a significant challenge for preprocessing.

- **Extremely High Missingness**: A group of nine variables (`id_07`, `id_08`, `id_21`, `id_22`, `id_23`, `id_24`, `id_25`, `id_26`, and `id_27`) exhibit missing value rates between 96.42% and 96.71%.
- **High Missingness**: Variables such as `id_18` (68.72% missing) and `id_03` and `id_04` (54.02% missing) also lack substantial amounts of data. `id_09`, `id_10`, and `id_33` hover around a 48-49% missing rate.

- **High Availability**: Conversely, `id_01` and `id_12` (alongside `TransactionID`) have complete availability (0% missing), while variables like `id_15` and `id_35` to `id_38` exhibit minimal missingness (approximately 2.25%).

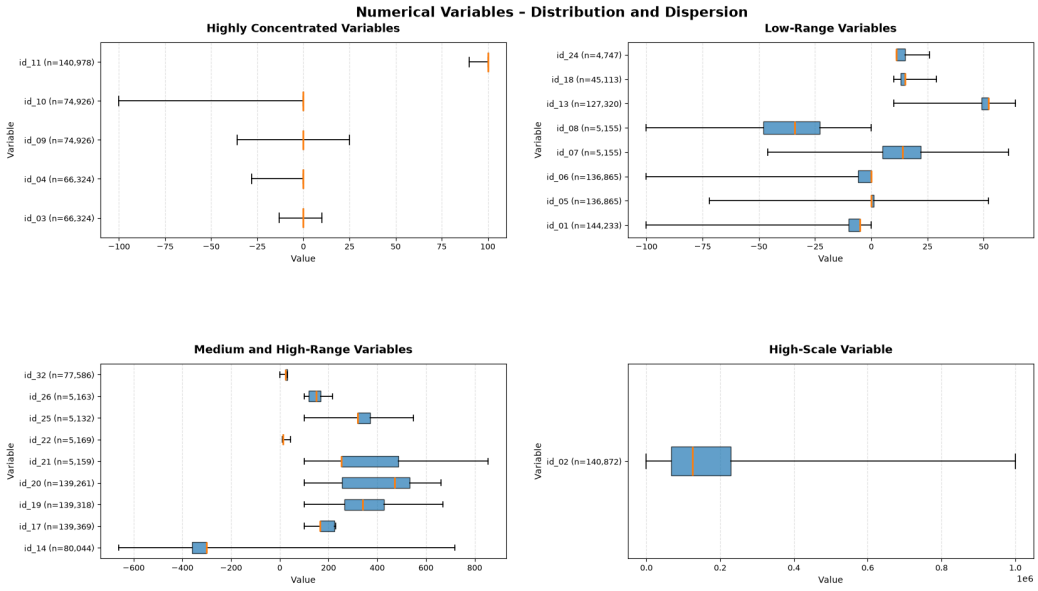


3.1.3 Numerical Variables: Distribution and Dispersion

The statistical summary of numerical features reveals significant heterogeneity in scale, variance, and behavior:

- **High Variance and Scale Disparities**: Features like `id_02` exhibit extreme dispersion, with a mean of 174,716.58 and a range spanning from 1 to 999,595. Variables `id_19` and `id_20` also display considerable variability. This highlights the potential need for normalization or standardization for distance-sensitive algorithms.
- **High Concentration (Pseudo-discrete behavior)**: Several `FLOAT64` variables are overwhelmingly concentrated on specific values, suggesting they act as discrete attributes or encoded categories. For instance, 94.46% of `id_11` observations equal 100. Similarly, `id_03`, `id_04`, `id_09`, and `id_10` have medians of 0, with zero accounting for 93.93% to 99.12% of their respective available data.

- **Extreme Values**: Significant gaps between minimum/maximum values and their respective quartiles suggest the presence of extreme values. For example, `id_01` has a minimum of -100 while its first quartile is -10, and `id_14` peaks at 720 despite a third quartile of -300. These are not necessarily errors but require careful evaluation as they may contain critical fraud-related signals.



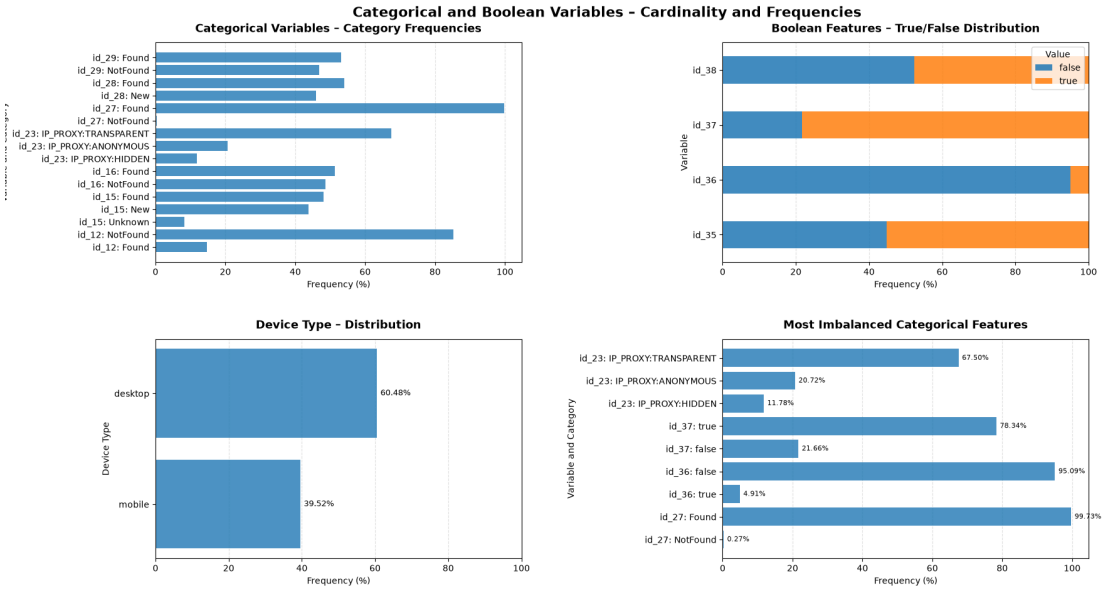
3.1.4 Categorical and Boolean Variables: Cardinality and Frequencies

The categorical variables exhibit varied levels of cardinality and highly imbalanced class distributions:

- **High Cardinality**: `DeviceInfo` possesses 1,786 distinct values (predominantly Windows, iOS Device, and MacOS). `id_30` (Operating Systems, mostly Windows 10/7) and `id_31` (Browsers, mainly Chrome and Safari) also show considerable diversity. `id_33` captures screen resolutions (e.g., 1920x1080), presenting an opportunity for feature engineering by splitting the string into numerical width and height dimensions.
- **Low Cardinality & High Concentration**: `id_27` is heavily skewed (99.73% 'Found'), as is `id_12` (85.30% 'NotFound') and `id_23` (67.50% 'IP_PROXY:TRANSPARENT'). The

variable `id_34` (`match_status`) is similarly imbalanced, with `status:2` accounting for 77.13% and `status:1` for 22.33%.

- **Boolean and Binary Features**: The `DeviceType` variable indicates a majority of desktop usage (60.48%) over mobile (39.52%). While `id_35` and `id_38` are fairly balanced, `id_36` is dominated by the value 'false' (95.09%) and `id_37` by 'true' (78.34%).



3.1.5 Correlation Analysis

A Pearson correlation analysis of the numerical variables indicates a general lack of strong linear relationships, reducing immediate concerns regarding multicollinearity. However, a few notable associations emerged:

- `id_17` and `id_22` demonstrate a positive correlation of 0.5920.
- `id_02` and `id_17` show a correlation of 0.4130.
- The most significant relationship is between `id_03` and `id_09` (0.7108). Because 94.80% of their shared observations are zeros, recalculating the correlation exclusively on non-zero

records raises the coefficient to 0.9193. However, this strong linear association must be interpreted cautiously, as it is derived from a much smaller subset of only 2,292 observations.



3.1.6 Summary of Implications for Preprocessing

The exploratory analysis of `identity_raw` confirms that it contains valuable, complex information regarding devices and digital identities. However, the subsequent preprocessing phase must address three major challenges:

1. **Missing Data Strategy**: The indiscriminate dropping of nulls would decimate the dataset. Instead, robust imputation strategies treating missingness as a strong predictive signal are required.
2. **Feature Engineering**: High-cardinality features (`id_30`, `id_31`, `DeviceInfo`) will require grouping or frequency encoding, and `id_33` should be parsed into numerical dimensions.
3. **Variable Transformations**: Given the stark differences in numerical scales and the discrete behavior of several `FLOAT64` variables, rigorous scaling techniques will be essential, particularly before applying topological data analysis (TDA) methods.

3.2 Transaction

The second phase of the Exploratory Data Analysis focused on the `transaction_raw` table. This table serves as the primary data source, containing economic metrics (`TransactionAmt`), temporal indicators (`TransactionDT`), categorical attributes (`ProductCD`, `card1`–`card6`, addresses, email domains), anonymized relational counts (`C1`–`C14`), time intervals (`D1`–`D15`), attribute matches (`M1`–`M9`), transformed features (`V1`–`V339`), and the target variable `isFraud`.

3.2.1 Dataset Structure and Data Types

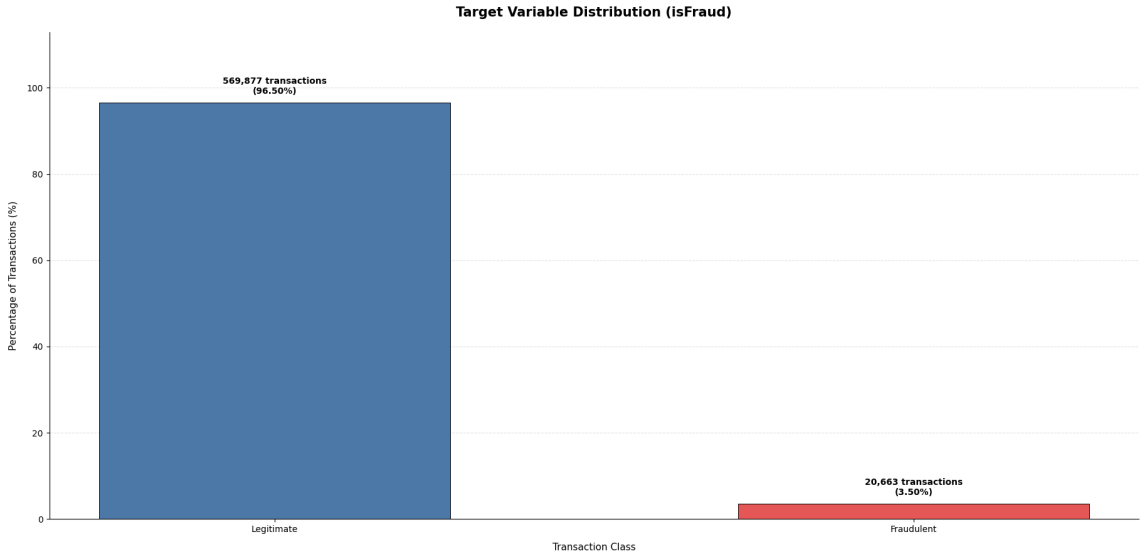
The `transaction_raw` dataset comprises 590,540 records and 394 variables. The data types are distributed as follows:

- **FLOAT64 (95.43%)**: The clear majority, representing 376 numerical attributes. Although stored as continuous floating-point numbers, many act as encoded categorical features or discrete counters.
- **BOOL (2.03%)**: Representing 8 binary indicators.
- **STRING (1.52%)**: Representing 6 categorical variables.
- **INT64 (1.02%)**: Representing 4 integer attributes, including the primary key `TransactionID` and the target variable `isFraud`.

Regarding data integrity, the primary identifier `TransactionID` contains no duplicate or missing values, ensuring each operation is uniquely identified across the dataset.

3.2.2 Target Variable Distribution and Class Imbalance

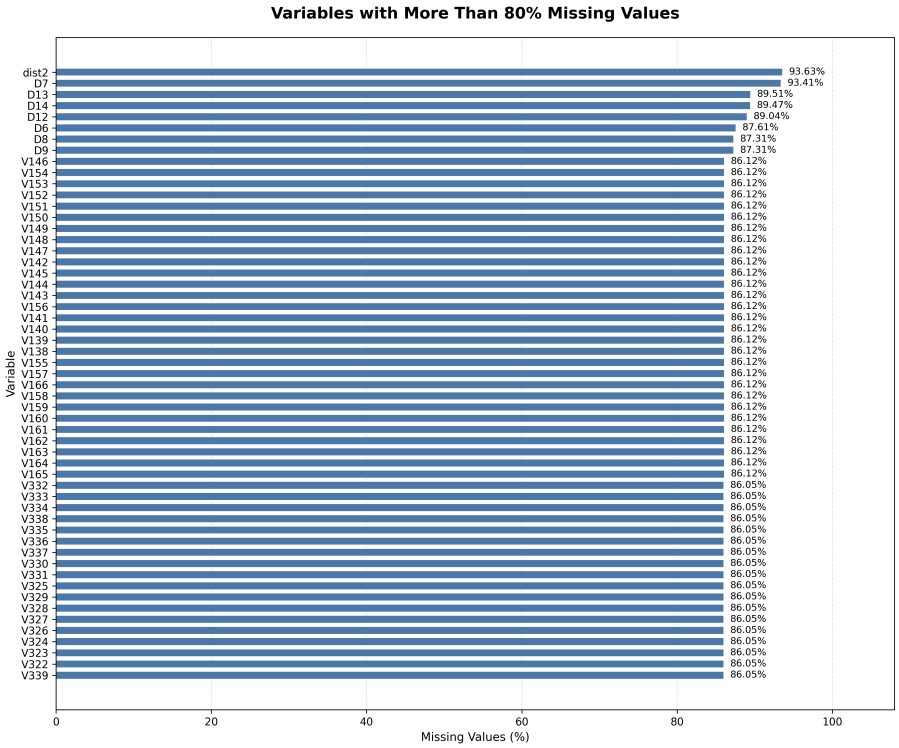
An analysis of the `isFraud` variable confirms a severe class imbalance, which is characteristic of fraud detection problems. Out of the 590,540 transactions, 569,877 are legitimate (96.50%), while only 20,663 are flagged as fraudulent (3.50%). This pronounced skewness necessitates the use of robust evaluation metrics (such as ROC-AUC or F1-score) during the modeling phase, as relying solely on accuracy would yield misleading results.



3.2.3 Missing Data Assessment

The presence of missing values is a defining challenge of this dataset, with a highly unequal distribution across variables:

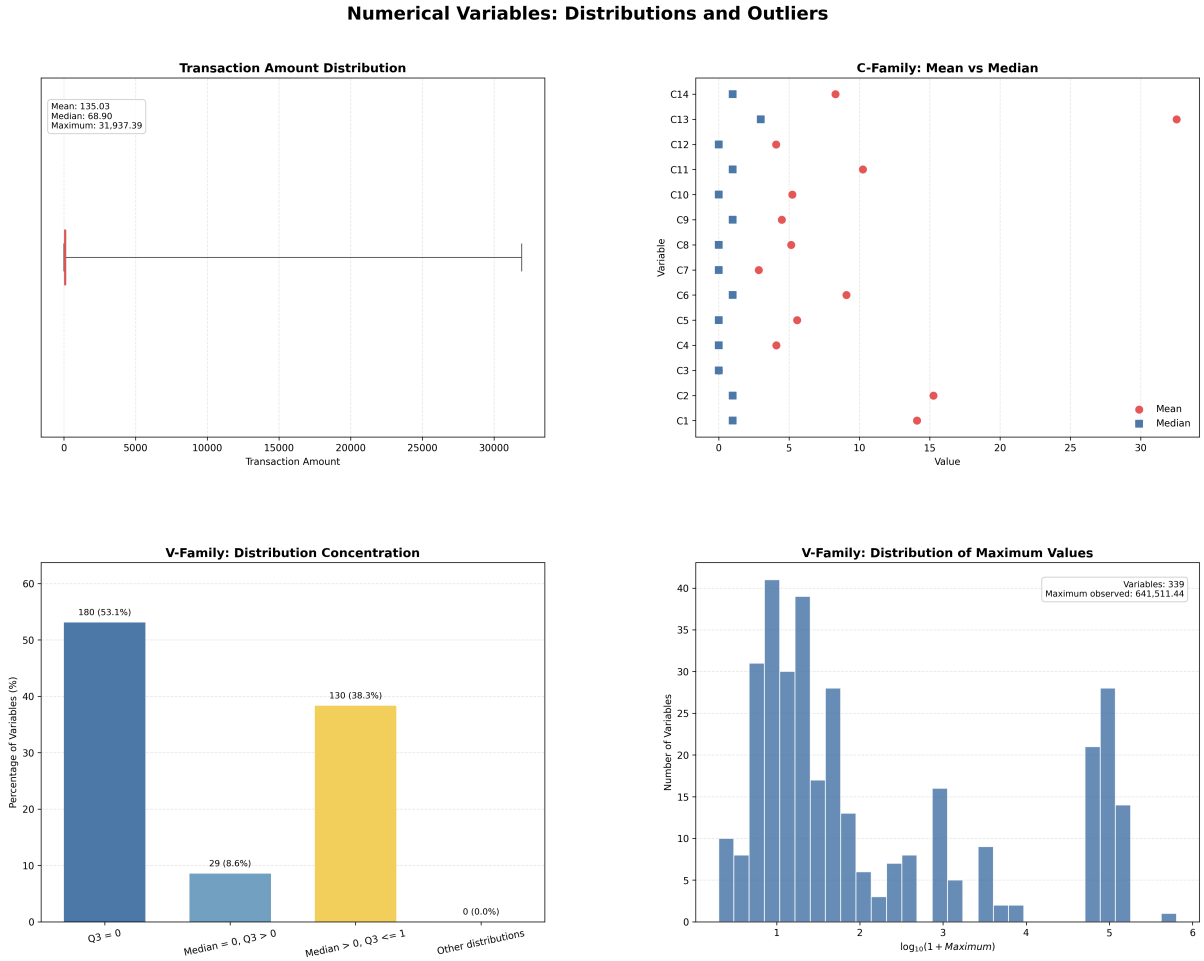
- **Complete Attributes**: Core variables such as `TransactionID`, `isFraud`, `TransactionDT`, `TransactionAmt`, and the `C1-C14` family are 100% complete.
- **High Missingness in Distance and Time**: Variables like `dist2` (93.63% missing) and `D7` (93.41% missing) lack data for the vast majority of transactions.
- **Categorical Missingness**: The recipient email domain (`R_emaildomain`) is missing in 76.75% of records, whereas the purchaser domain (`P_emaildomain`) is missing in only 15.99%.
- **Anonymized Blocks**: The `V` family features exhibit shared missingness patterns. For instance, variables `V138` to `V166` and `V322` to `V339` show null rates approaching 86%.



3.2.4 Numerical Variables: Distributions and Outliers

A descriptive statistical analysis of the numerical variables highlights significant asymmetry and the presence of extreme values:

- **Transaction Amount (`TransactionAmt`)**: This variable is heavily right-skewed. The mean is 135.03, while the median is 68.90, and the maximum value reaches 31,937.39. Using the Interquartile Range (IQR) method, 66,480 observations (11.26%) were identified as statistical outliers. A logarithmic transformation may be necessary for distance-based models.
- **C-Family Counters**: Features `C1` through `C14` are highly asymmetric. For instance, `C1` has a median of 1 but a mean of 14.09, driven by extreme maximum values.
- **V-Family Characteristics**: Many `V` variables exhibit pseudo-discrete behavior, heavily concentrated on zeros and ones. Their scales vary drastically, with some variables bounded at low integers and others exceeding magnitudes of 100,000, underscoring the need for rigorous scaling techniques.



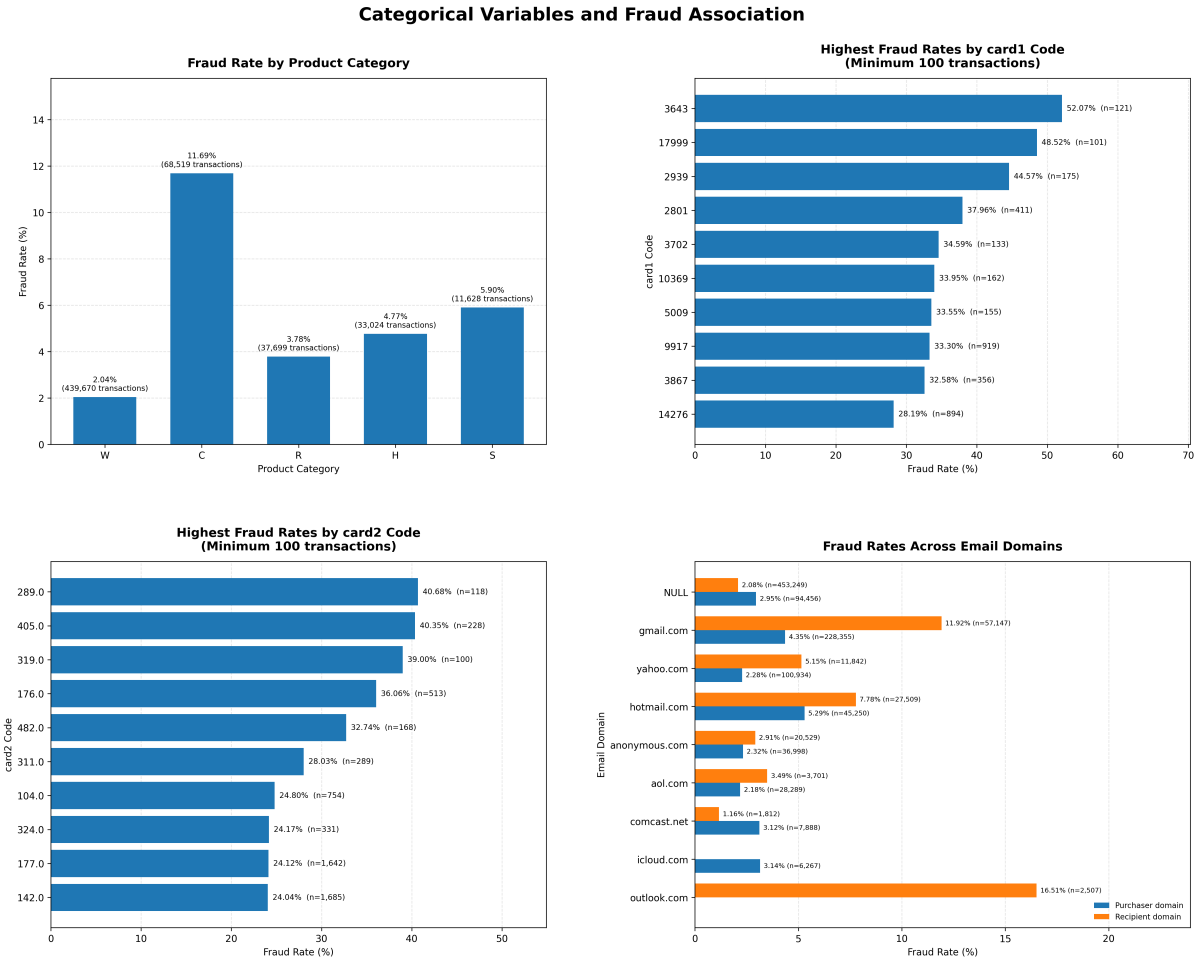
3.2.5 Categorical Variables and Fraud Association

An evaluation of categorical variables reveals significant disparities in fraud rates across different segments:

- **Product Category** (`ProductCD`): Category 'W' dominates the dataset (74.45%) but has a low fraud rate of 2.04%. Conversely, category 'C', representing only 11.60% of transactions, accounts for a disproportionate amount of fraud, with an 11.69% fraud rate.
- **Card Features**: Credit transactions display a significantly higher fraud risk (6.68%) compared to debit transactions (2.43%) within `card6`. Additionally, `card1` (13,553 distinct

values) and `card2` (500 distinct values) possess high cardinality. Despite this, specific codes within these variables exhibit fraud rates exceeding 30% or 40% across statistically significant sample sizes, highlighting their predictive potential.

- **Email Domains**: While `gmail.com` is the most frequent domain in both `P_emaildomain` and `R_emaildomain`, fraud rates fluctuate across providers. For example, `P_emaildomain` transactions linked to `hotmail.com` show a 5.30% fraud rate, compared to 2.28% for `yahoo.com`.



3.2.6 Correlation Analysis

To evaluate the linear relationship between numerical/boolean features and the target variable, a Pearson correlation analysis was conducted:

- **C, D, and M Families**: These variables generally exhibit weak linear correlations with `isFraud`. The strongest negative associations are found in `D8` (-0.1426) and `D7` (-0.1272). Within the `M` family, after converting boolean values to numerical formats, category `M2` within the `M4` feature stands out with a positive correlation of 0.1439.
- **V Family**: Unlike the previous groups, several anonymized `V` variables demonstrate moderate positive linear associations with fraud. The most notable are `V257` (0.3831), `V246` (0.3669), `V244` (0.3641), and `V242` (0.3606).



It is crucial to note that weak linear correlations (such as those observed with `TransactionAmt` or `dist1`) do not imply a lack of predictive value, as non-linear relationships or complex interactions may still exist.

3.2.7 Summary of Implications for Preprocessing

The EDA of the `transaction_raw` dataset dictates several critical preprocessing steps:

1. **Class Imbalance Mitigation**: Intelligent undersampling of the majority class must be applied to prevent the machine learning models from biasing toward legitimate transactions.
2. **Missing Data Management**: Because missingness patterns inherently correlate with fraud, simple imputation is insufficient. Models capable of natively utilizing sparsity as a predictive signal are required.
3. **High Cardinality Encoding**: Variables like `card1`, `card2`, and email domains require advanced categorical encoding strategies (e.g., Frequency Encoding) to preserve their valuable informational signal without drastically expanding the overall feature space.

Chapter 4

Data Preprocessing and Transformation

This chapter details the data preparation and feature engineering pipeline designed to transform raw datasets into structured feature matrices optimized for machine learning models. The preprocessing methodology is split into two distinct experimental tracks: a traditional domain-driven feature engineering track (**Baseline**) and an advanced topological track (**TDA-Enriched**) that incorporates geometric persistent homology properties to capture complex fraudulent structures.

To ensure strict reproducibility and auditability, all code artifacts, validation scripts, and execution outputs have been made publicly accessible in the project's GitHub repository. The repository contains the modular PySpark scripts (`.py`) executed for each transformation stage, the BigQuery SQL verification queries (`.sql`) used to validate table views and data integrity between steps, their corresponding execution outputs in JSON format (`.json`), and visual execution validation snapshots (`.png`).

To address the limitations of raw transactional inputs and provide gradient-boosted decision trees (GBDT) with strong predictive signals, a domain-specific feature engineering pipeline was implemented. The baseline workflow begins by consolidating the raw `transaction` and `identity` datasets via an initial `LEFT JOIN` on `TransactionID`.

4.1 Baseline Feature Engineering

Baseline track executes a sequential 6-stage transformation pipeline:

1. `etl_01_temporal.py`: Decodes continuous timestamps into cyclical diurnal and weekly time indicators.
2. `etl_02_uid.py`: Synthesizes virtual user keys (`UID_card_addr`) and computes historical activity density.
3. `etl_03_email.py`: Standardizes raw email domains into corporate parent providers and extracts domain mismatch flags.
4. `etl_04_user_aggregations.py`: Computes relative spend deviations (μ_{UID}) and amount velocity ratios.
5. `etl_05_encoding_selection.py`: Applies null-preserving label encoding and purges raw string identifiers.
6. `etl_06_train_test_baseline.py`: Enforces out-of-time (OOT) temporal partitioning and calculates class imbalance weights.

The following subsections detail the rationale, mathematical formulation, and empirical validation for each of these baseline engineering phases.

4.1.1 Phase 1: Temporal Feature Engineering

Rationale & Utility: Raw timestamps (`TransactionDT`) measure continuous elapsed seconds from an arbitrary origin, making them unsuitable for tree splits and prone to overfitting. Decoding them into cyclical diurnal and weekly components allows XGBoost to isolate non-linear risk windows and off-peak fraud spikes without temporal data leakage.

Continuous timestamps were transformed into discrete time indicators:

$$\text{DT_hour} = \left\lfloor \frac{\text{TransactionDT}}{3600} \right\rfloor \pmod{24} \tag{4.1}$$

$$\text{DT_day_of_week} = \left\lfloor \frac{\text{TransactionDT}}{86400} \right\rfloor \pmod{7} \tag{4.2}$$

$$\text{DT_day_continuous} = \left\lfloor \frac{\text{TransactionDT}}{86400} \right\rfloor \tag{4.3}$$

Validation across the 182-day scope ($N = 590,540$) confirms a pronounced non-linear relationship between time of day and fraud probability (Table 4.1).

Table 4.1: Distribution of Transactions and Fraud Rates by Hour of Day (`DT_hour`)

Time Window / Hour	Total Volume	Fraud Count	Fraud Rate (%)	Risk Level
05:00 – 07:59	19,412	1,542	7.94% – 10.61%	**Extreme**
08:00 – 09:59	5,070	464	9.00% – 9.30%	**High**
00:00 – 04:59	142,965	4,809	3.13% – 5.19%	Moderate
10:00 – 12:59	22,905	837	3.04% – 5.32%	Moderate
13:00 – 15:59	82,502	2,011	2.29% – 2.54%	Low (Baseline)
16:00 – 23:59	317,686	11,000	2.95% – 3.70%	Baseline

Key Insights: Off-peak hours (`05:00`–`09:59`) present extreme fraud concentration, peaking at **10.61% at 07:00 AM** (over 3× the population average of $\sim$ 3.5%). Conversely, peak business volume (`13:00`–`15:59`) exhibits the lowest risk (2.29%), establishing `DT_hour` as a primary split candidate.

4.1.2 Phase 2: Synthetic Virtual User Identifiers (UIDs)

Rationale & Utility: The dataset lacks a explicit customer account ID. Synthesizing a Virtual User Identifier (`UID_card_addr`) enables entity disambiguation, allowing the model to track individual user behavior, measure transaction velocity, and detect high-frequency card testing attacks.

The synthetic key concatenates static card attributes and location metadata:

$$\text{UID_card_addr} = \text{card1} \parallel \text{card2} \parallel \text{card3} \parallel \text{card4} \parallel \text{card5} \parallel \text{card6} \parallel \text{addr1} \tag{4.4}$$

Frequency encoding (`UID_card_addr_fq`) was computed to quantify activity density across the 43,018 synthesized entities.

Table 4.2: Transaction Distribution and Fraud Concentration by User Activity Bucket

User Activity Bucket	Total Volume	Fraud Count	Fraud Rate (%)	% of Total Fraud
1 Transaction (Single-use)	17,358	572	3.30%	2.77%
2 – 5 Transactions	40,121	1,216	3.03%	5.88%
6 – 20 Transactions	80,177	2,246	2.80%	10.87%
> 20 Transactions (High Volume)	452,884	16,629	3.67%	**80.48%**
Total / Summary	**590,540**	**20,663**	**3.50%**	**100.00%**

Key Insights: Resolving 590,540 records into 43,018 behavioral profiles reveals that high-velocity entities (> 20 transactions) account for **80.48%** of all fraud instances, equipping XGBoost with an explicit signal for automated attacks.

4.1.3 Phase 3: Email Domain Standardization & Matching

Rationale & Utility: Raw purchaser (`P_emaildomain`) and recipient (`R_emaildomain`) email fields exhibit high cardinality and sparsity. Standardizing these inputs into corporate parent providers prevents feature space explosion, while extracting domain mismatch flags captures identity deception in cross-border transactions.

Regex parsing mapped domain suffixes into 8 major categories, and a binary match feature was generated:

$$\text{email_domain_match} = \begin{cases} 1 & \text{if P_emaildomain} = \text{R_emaildomain} \neq \varnothing \\ 0 & \text{otherwise} \end{cases} \tag{4.5}$$

Table 4.3: Transaction Distribution and Fraud Rates by Mapped Purchaser Email Provider

Mapped Provider (`P_emaildomain_bin`)	**Total Volume**	**Fraud Count**	**Fraud Rate (%)**	**% of Total Fraud**
Microsoft	59,477	3,170	**5.33%**	15.34%
Google	228,851	9,954	**4.35%**	**48.17%**
Missing	94,456	2,790	2.95%	13.50%
Apple	8,225	238	2.89%	1.15%
Anonymous	36,998	859	2.32%	4.16%
AOL	28,604	657	2.30%	3.18%
Yahoo	105,305	2,370	2.25%	11.47%
Other	28,624	625	2.18%	3.03%
Total / Summary	**590,540**	**20,663**	**3.50%**	**100.00%**

Key Insights: Categorical cardinality was reduced by $> 86\%$. Risk asymmetry is evident: `Microsoft` domains exhibit a 5.33% fraud rate (2.44× higher than legacy providers), while `Google` and `Microsoft` combined concentrate **63.51%** of total fraud events.

4.1.4 Phase 4: Relative Amount Velocity Aggregations

Rationale & Utility: Absolute transaction amounts (`TransactionAmt`) are uninformative without user context. Normalizing spend against a user's historical baseline isolates sudden purchasing anomalies and account takeover spikes, distinguishing high-spending legitimate users from compromised low-tier accounts.

Aggregations were computed over each `UID_card_addr`:

$$\text{uid_amt_diff_mean} = \text{TransactionAmt} - \mu_{\text{UID}} \tag{4.6}$$

$$\text{uid_amt_ratio_mean} = \frac{\text{TransactionAmt}}{\mu_{\text{UID}} + \epsilon} \tag{4.7}$$

where $\mu_{\text{UID}} = \mathbb{E}[\text{TransactionAmt} \mid \text{UID_card_addr}]$ and $\epsilon = 10^{-5}$ handles single-transaction profiles.

Table 4.4: Transaction Volume and Fraud Concentration by Relative Amount Deviation Bucket

Amount Deviation Bucket	Total Volume	Fraud Count	Fraud Rate (%)	% of Total Fraud
1.5x – 3x User Average (Moderate Spike)	67,373	3,471	**5.15%**	16.80%
> 3x User Average (Extreme Spike)	23,383	1,076	**4.60%**	5.21%
Normal Range (∼1x User Average)	176,027	6,720	3.82%	32.52%
< 0.8x User Average (Below Usual)	323,757	9,396	2.90%	45.47%
Total / Summary	**590,540**	**20,663**	**3.50%**	**100.00%**

Key Insights: Spending spikes between 1.5× and 3.0× a user's average yield the highest fraud incidence (**5.15%**, a 1.47× lift over baseline), confirming relative velocity as a critical anomaly metric.

4.1.5 Phase 5: Matrix Encoding & Standardization

Rationale & Utility: GBDT algorithms require numerical matrices. To prevent severe dimensionality expansion from One-Hot Encoding on high-cardinality strings, null-preserving Label Encoding (`StringIndexer`) was implemented, preserving native missing-value split paths in XGBoost.

Uninformative keys (`TransactionID`, unparsed emails, raw strings) were purged, resulting in a clean numerical feature matrix $\mathbf{X} \in \mathbb{R}^{590540 \times 443}$ (Table 4.5).

Table 4.5: Feature Matrix Pipeline Schema Evolution Across ETL Phases

ETL Phase / Storage Artifact	Total Records (N)	Column Count	String Attributes
Phase 0 (Raw Joined Parquet)	590,540	434	31
Phase 1 (`step_01_temporal.parquet`)	590,540	437	31
Phase 2 (`step_02_uid.parquet`)	590,540	439	32
Phase 3 (`step_03_email.parquet`)	590,540	442	34
Phase 4 (`step_04_agg.parquet`)	590,540	448	34
Phase 5 (`step_05_encoding_selection.parquet`)	**590,540**	**444**	**0**

4.1.6 Phase 6: Temporal Partitioning & Imbalance Calibration

Rationale & Utility: Random k-fold splitting introduces temporal data leakage and optimistic bias. Enforcing a strict Out-Of-Time (OOT) temporal split simulates real operational deployment. Furthermore, calibrating the GBDT loss function directly addresses the severe class imbalance without synthesizing artificial data via SMOTE.

Transactions from Days 1–145 were allocated to training (82.10%), while Days 146–182 were reserved for OOT evaluation (17.90%). Target prevalence remained stable (3.52% train vs. 3.42% test).

To handle the $1 : 27.44$ positive-to-negative ratio, the gradient loss parameter `scale_pos_weight` was calibrated on the training partition:

$$\text{scale_pos_weight} = \frac{N_{\text{train, negative}}}{N_{\text{train, positive}}} = \frac{467,795}{17,052} \approx 27.43 \tag{4.8}$$

This forces XGBoost to scale positive gradients, directly optimizing PR-AUC during model convergence.

4.2 TDA-Enriched Feature Engineering

To evaluate whether geometric properties derived from persistent homology can provide a first-order predictive signal beyond classical tabular features, an enriched experimental track was constructed. The **TDA-Enriched** pipeline shares the foundational feature engineering steps executed in the baseline track (Phases 1 through 4), but branches directly after user aggregations to introduce a dedicated topological extraction stage.

The complete TDA-Enriched workflow follows a sequential 7-stage processing structure:

1. `etl_01_temporal.py` to `etl_04_user_aggregations.py`: Shared foundational feature engineering steps (temporal decoding, UID synthesis, email standardization, and amount velocity ratios).
2. `etl_05_tda.py`: Extracts 1-dimensional persistent homology properties, computing Euclidean distances to the top 5 H_1 topological loops.
3. `etl_06_encoding_selection.py`: Standardizes residual categorical features via null-preserving label encoding and formats the enriched feature matrix.
4. `etl_07_train_test_tda_enriched.py`: Enforces strict out-of-time (OOT) temporal partitioning and calculates class imbalance weights for the expanded feature matrix.

4.2.1 Phase 5: Topological Feature Extraction

For readers who may not yet be fully familiar with the mathematical foundations of Topological Data Analysis, the formal definitions introduced in this chapter can initially appear abstract. To build a more solid and intuitive understanding of concepts such as simplicial complexes, filtrations, and persistent homology, a practical introductory guide is provided in my previous Bachelor's thesis, which was specifically written to bridge the gap between algebraic topology and applied data science.

Rationale & Utility: Traditional tabular features evaluate transactions in isolation or against static aggregations. However, financial fraud frequently operates in coordinated patterns (e.g., fraud rings, card testing loops, automated bot networks) that form structural voids in high-dimensional feature space.

Geometric Intuition: Hyperplanes vs. Topological Boundaries

To fully grasp the predictive value of Topological Data Analysis, it is essential to understand the geometric mechanics of gradient-boosted decision trees (GBDT) like XGBoost. Tree-based models partition the feature space using orthogonal, axis-parallel hyperplanes. Mathematically, every decision split evaluates a single feature against a scalar threshold (e.g., $X_1 > \theta$), effectively drawing a straight, grid-like boundary across the multidimensional space.

While highly efficient for tabular data, this orthogonal partitioning struggles against the complex geometry of organized fraud. Fraudulent networks often exhibit cyclical behaviors that manifest as non-linear structures—specifically, topological loops or rings. When an algorithm attempts to enclose a circular or ring-like cluster using only straight, intersecting hyperplanes, it inevitably creates a "step-function" boundary. This rigid boundary inadvertently captures surrounding areas of the feature space that belong to legitimate transactions, generating dense pockets of False Positives.

The intuitive power of TDA lies in transforming this complex non-linear geometry into a linear problem that XGBoost can easily solve. By locating the center (centroid) of a fraudulent topological loop and calculating the continuous Euclidean distance from every transaction to that center, the spatial complexity is compressed into a single 1-dimensional feature.

Consequently, when XGBoost makes a simple orthogonal split on this new TDA feature (e.g., dist_to_h1_loop $<$ radius), it is mathematically equivalent to drawing a perfect, non-linear hyper-spherical boundary around the fraud ring in the original feature space. This empowers the tree model to isolate complex fraud structures with surgical precision, drastically reducing False Positives.

Dimensionality Reduction and Stratified Subsampling

To understand the absolute necessity of dimensionality reduction and subsampling in this phase, one must consider the theoretical computational complexity inherent to TDA. Computing persistent homology relies on constructing a filtration of simplicial complexes (typically the Vietoris-Rips complex) across an increasing proximity radius ϵ. As ϵ grows, data points intersect to form k-simplices. The maximum number of k-simplices that can be formed from N data points is governed by the binomial coefficient $\binom{N}{k+1}$, scaling at an order of $\mathcal{O}(N^{k+1})$. Consequently, the total number of simplices S in the worst-case scenario grows exponentially to $\mathcal{O}(2^N)$.

Furthermore, the standard algebraic algorithm to compute persistent homology (boundary matrix reduction) has a worst-case computational complexity of $\mathcal{O}(S^3)$. When evaluating the full dataset of $N = 590,540$ transactions across the baseline dimension of $D = 444$, calculating just the initial pairwise distance matrix requires $\mathcal{O}(N^2 \cdot D)$ operations. Constructing the subsequent Vietoris-Rips complex would result in an intractable combinatorial explosion, instantly exhausting computational resources.

To make the topological extraction computationally viable, the pipeline explicitly leverages the results previously generated by the trained Baseline model. First, the feature space was restricted to the **Top 20 most important variables**, as determined by the Average Gain metric from the baseline XGBoost model. By projecting the data into this reduced 20-dimensional space, the computational cost of the distance matrix is drastically reduced, and the point cloud focuses strictly on the most predictive structural relationships without topological noise.

Second, a representative subsample of 4,000 transactions was drawn directly from the baseline model's out-of-time validation predictions. This subsample size was analytically chosen to restrict the simplex generation S to a computationally stable boundary. The sample was stratified based on the baseline model's classification error distribution. As observed in the sample error breakdown, False Positives $(2,123)$ heavily dominated the sample, representing specific areas where the baseline XGBoost model failed geometrically. By computing the topological loops precisely on this error-dense subsample within the Top 20 predictive dimensions, the TDA extraction bypasses the theoretical $\mathcal{O}(S^3)$ bottleneck and focuses exclusively on mapping the complex boundaries where traditional decision trees struggle.

Persistent Homology Computation

Persistent homology was computed using the Ripser library on the reduced 20-dimensional dataset. The algorithm evaluates the simplicial complexes across increasing filtration scales to identify persistent topological structures. The resulting Persistence Diagram visualizes these features (see Figure 4.1). In this representation, the H_0 features (blue dots) correspond to

isolated connected components, while the H_1 features (orange dots) represent the topological loops. Features located furthest from the diagonal represent the most persistent and structurally significant geometric shapes within the transaction data, distinguishing true structural voids from mere topological noise.

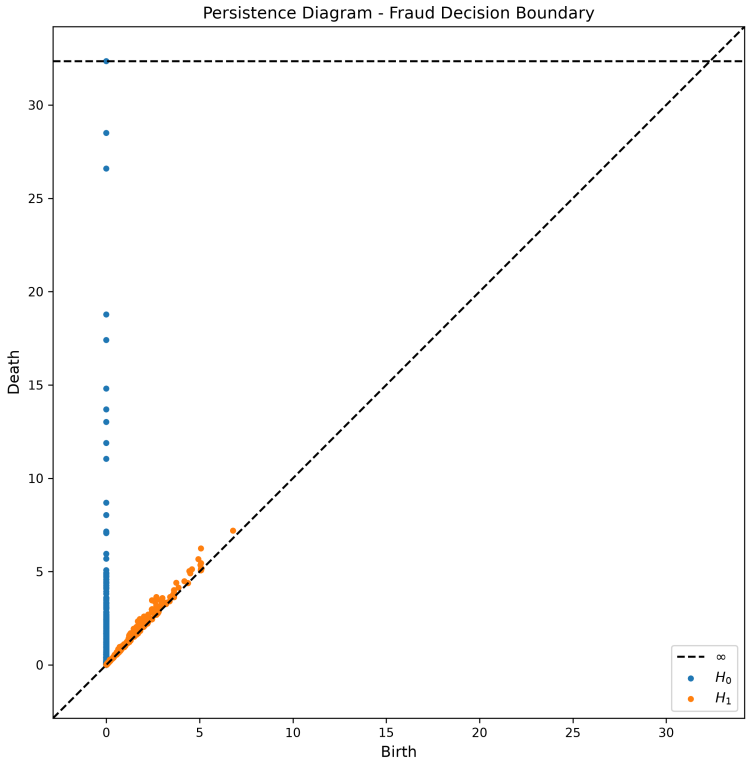


Figure 4.1: Persistence Diagram representing H_0 (connected components) and H_1 (topological loops) across filtration thresholds.

Inverse Mapping and Geometric Error Analysis

Once the persistent topological loops were mathematically identified, an inverse mapping process was executed to trace these geometric structures back to their original data points. The pipeline isolated the top 5 most persistent H_1 features to analyze their internal composition. For example, Loop #1 (Persistence: 1.165) involved exactly 113 specific transactions.

Crucially, the classification error breakdown within this single topological loop contained a complex mix of 73 True Negatives and 26 False Positives. This illustrates exactly how fraudulent and legitimate transactions intertwine in non-linear loop structures that a simple orthogonal decision split cannot cleanly separate.

This geometric failure of classical gradient boosting is clearly visualized in Figure 4.2, which projects the feature space using Principal Component Analysis (PCA). The dashed vertical line represents a standard orthogonal XGBoost decision boundary. Because tree-based models rely on axis-parallel splits, the model is forced to draw a rigid line that incorrectly classifies a massive zone of transactions as False Positives (represented by the red dots inside the shaded classification error zone). By contrast, the true topological loop (indicated by the solid circle and its calculated centroid) captures the complex, non-linear relationship of the data points, enclosing the fraud structure perfectly where orthogonal splits fail.

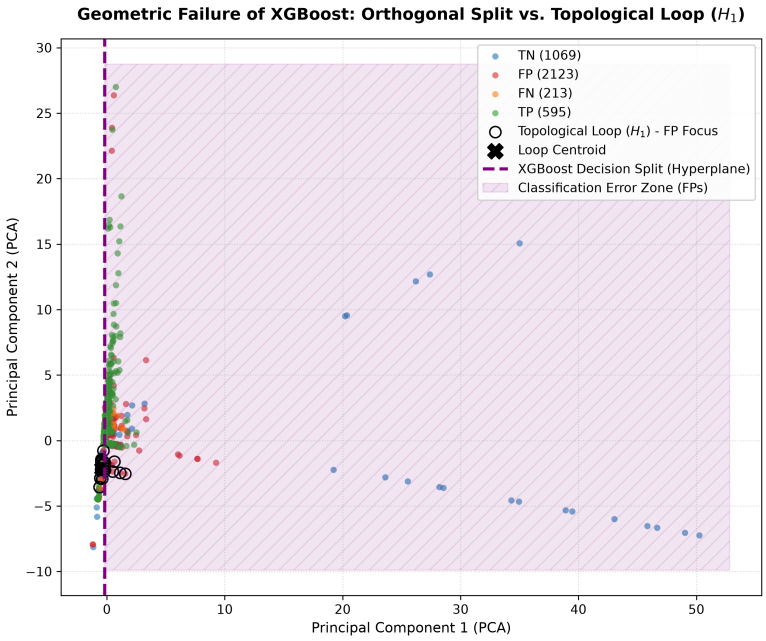


Figure 4.2: PCA projection demonstrating the geometric failure of orthogonal XGBoost splits versus the non-linear boundary of a Topological Loop (H_1).

Feature Generation: Distances to H_1 Centroids

After identifying the precise transactions forming the top 5 most persistent H_1 loops, the spatial centroid of each loop was calculated in the 20-dimensional space. These centroids were exported as metadata (`tda_centroids.json`) and uploaded to Google Cloud Storage.

Finally, for every transaction in the full dataset, the minimum Euclidean distance from its Top 20 feature vector to each of the 5 spatial centroids was computed. Let $\mathbf{x}_i \in \mathbb{R}^{20}$ represent the

restricted feature vector of transaction i, and let $\mathbf{c}_k$ denote the calculated centroid of persistent homology loop $k \in \{1, 2, 3, 4, 5\}$ in H_1. The new topological distance feature dist_to_h1_loop_k is formally defined as:

$$\text{dist_to_h1_loop_}k(\mathbf{x}_i) = \|\mathbf{x}_i - \mathbf{c}_k\|_2 \tag{4.9}$$

This transformation successfully appends 5 continuous geometric features to the dataset:

$$\begin{aligned}\mathcal{F}_{\text{TDA}} = \{&\text{dist_to_h1_loop_1, dist_to_h1_loop_2, dist_to_h1_loop_3,} \\ &\text{dist_to_h1_loop_4, dist_to_h1_loop_5}\}\end{aligned} \tag{4.10}$$

Chapter 5

Modeling

To evaluate the predictive power of classical transactional features against variables derived from Topological Data Analysis (TDA), **eXtreme Gradient Boosting (XGBoost)** was selected as the core analytical engine. Its selection addresses the specific complexities of the dataset through four fundamental capabilities:

- **Native Missing Value Handling:** It employs a sparsity-aware split-finding algorithm that natively learns the optimal default direction for missing data (Null/NaN). This avoids the artificial biases introduced by explicit imputation strategies required by traditional algorithms.
- **Intrinsic Feature Importance:** XGBoost provides transparent metrics based on information gain. This interpretability is essential for our core objective of extracting a clear, objective ranking to compare novel TDA variables against classical features.
- **Robustness to Multicollinearity:** Tree-based ensembles naturally handle highly correlated spatial and topological features. By tuning hyperparameters like `colsample_bytree`, the model is forced to evaluate subsets of variables, ensuring TDA signals are explored even if dominant classical features are present.
- **Class Imbalance Management:** The native `scale_pos_weight` hyperparameter explicitly penalizes minority class misclassifications, effectively addressing the severe fraud imbalance ($< 4\%$) without synthesizing artificial data.

By establishing XGBoost as a robust baseline, we create a transparent framework to quantify TDA's added value. To ensure full reproducibility and operational transparency, all modeling artifacts are publicly accessible in the project's GitHub repository. This includes the modular

Python scripts (`.py`) used for distributed model training, the execution Bash scripts (`.sh`) designed for automated orchestration via Google Cloud Shell, and the resulting performance validation plots and feature importance graphs (`.png`).

5.1 Baseline Model Results and Evaluation

Following the execution of the baseline feature engineering pipeline, the XGBoost model was trained and evaluated on the out-of-time test partition. The performance analysis relies on global discrimination metrics, intrinsic feature importance, and the probability density of the predictions.

5.1.1 Global Performance Metrics

The baseline model demonstrates a strong capacity to distinguish between legitimate and fraudulent transactions. As detailed in the evaluation curves, the model achieved a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.9128. Furthermore, it yielded a Precision-Recall Area Under the Curve (PR-AUC) of 0.5300. In the context of highly imbalanced financial datasets, maintaining a PR-AUC above the 0.50 threshold indicates a robust precision-to-recall tradeoff.

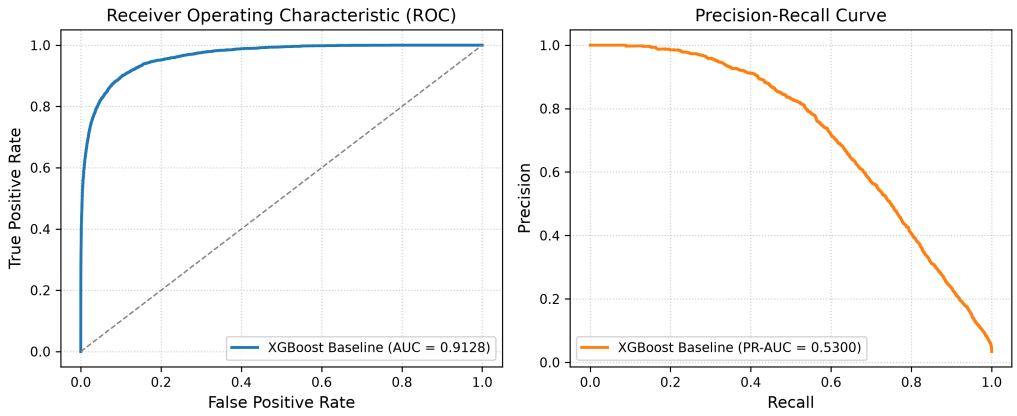


Figure 5.1: ROC and Precision-Recall curves for the Baseline XGBoost model.

5.1.2 Feature Importance

The intrinsic feature selection mechanism of XGBoost highlights the predictive dominance of specific tabular variables. Based on the Average Gain metric, the top three most important features are `V258`, `V218`, and `V264`. Notably, the variable `V258` exhibits the highest average gain by a significant margin, outperforming all other variables in the feature space.

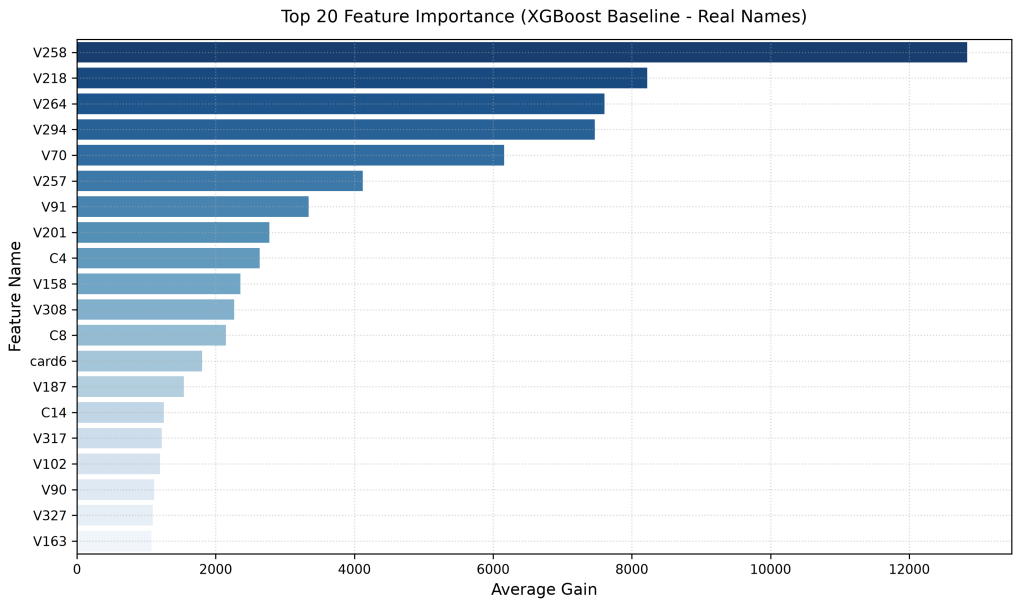


Figure 5.2: Top 20 most important features in the Baseline model ranked by Average Gain.

5.1.3 Classification Error and Probability Density

When applying the default probability decision boundary, the model successfully identified 2,709 True Positives and 92,361 True Negatives. However, this threshold generated a substantial volume of False Positives, misclassifying 9,721 legitimate transactions as fraud, alongside 902 False Negatives.

This classification error is visually explained by the probability distribution of the predictions. While the actual legitimate class (0) is heavily concentrated near zero, its right tail extends beyond the 0.5 decision boundary. Conversely, the predicted probabilities for the actual fraud class (1) are broadly distributed across the spectrum, with a significant portion correctly falling above the 0.5 threshold.

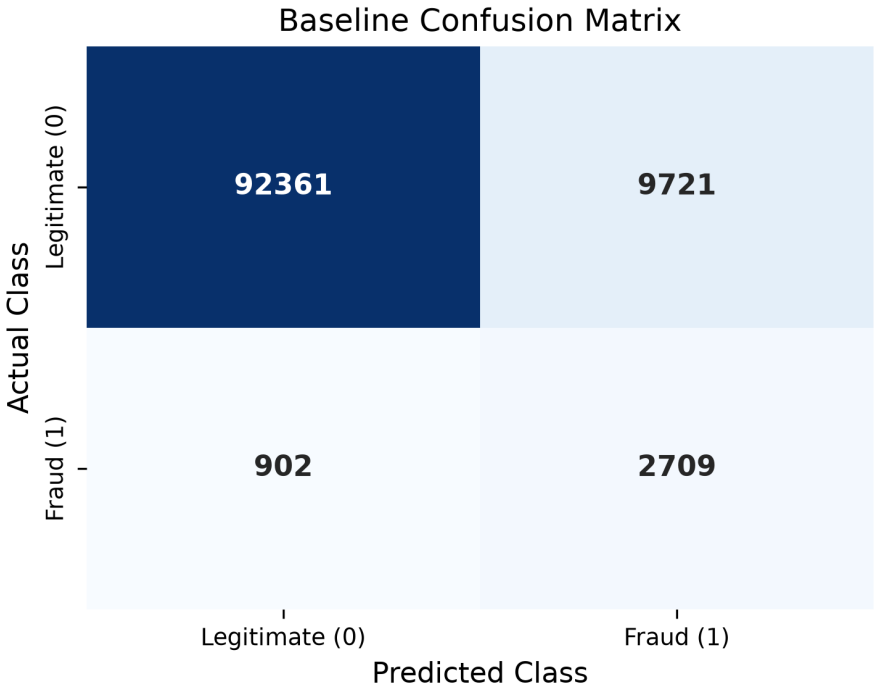


Figure 5.3: Baseline Confusion Matrix at 0.5 threshold.

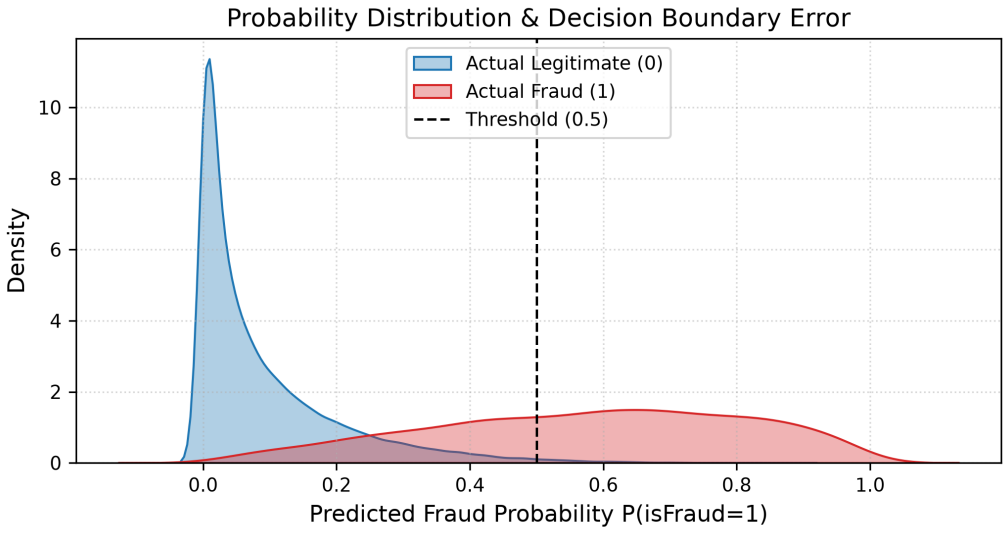


Figure 5.4: Probability density distribution for legitimate and fraud classes.

5.2 TDA-Enriched Model Results and Evaluation

Following the integration of the persistent homology features into the feature space, the TDA-enriched XGBoost model was trained and evaluated under the same out-of-time temporal partition. This section examines the impact of topological enrichment on global model discrimination, feature importance hierarchies, and classification error distribution.

5.2.1 Global Performance Metrics

The TDA-enriched model achieves a slight improvement in global discriminative capability compared to the baseline. As shown in the evaluation curves, the model yields a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.9144 (up from 0.9128). The Precision-Recall Area Under the Curve (PR-AUC) registers at 0.5272, maintaining a stable precision-to-recall tradeoff while incorporating complex geometric spatial dimensions.

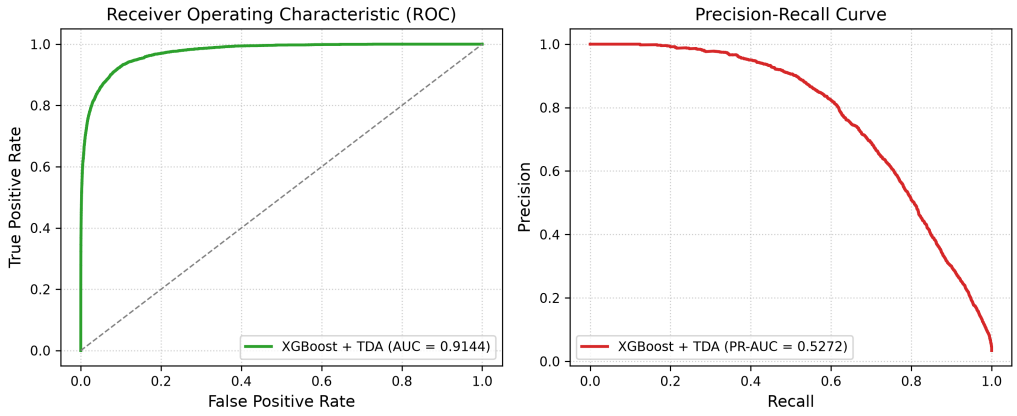


Figure 5.5: ROC and Precision-Recall curves for the TDA-Enriched XGBoost model.

5.2.2 Feature Importance and Topological Dominance

One of the most significant findings of this study is revealed by the feature importance ranking based on Average Gain. As illustrated in Figure 5.6, the newly engineered topological features–specifically `dist_to_h1_loop_4`, `dist_to_h1_loop_5`, and `dist_to_h1_loop_3`–immediately dominate the top tiers of the feature importance hierarchy, outperforming traditional tabular variables like `V258` and `V295`. This confirms that the geometric representation of persistent homology loops captures critical non-linear variance that classical tabular features fail to express natively.

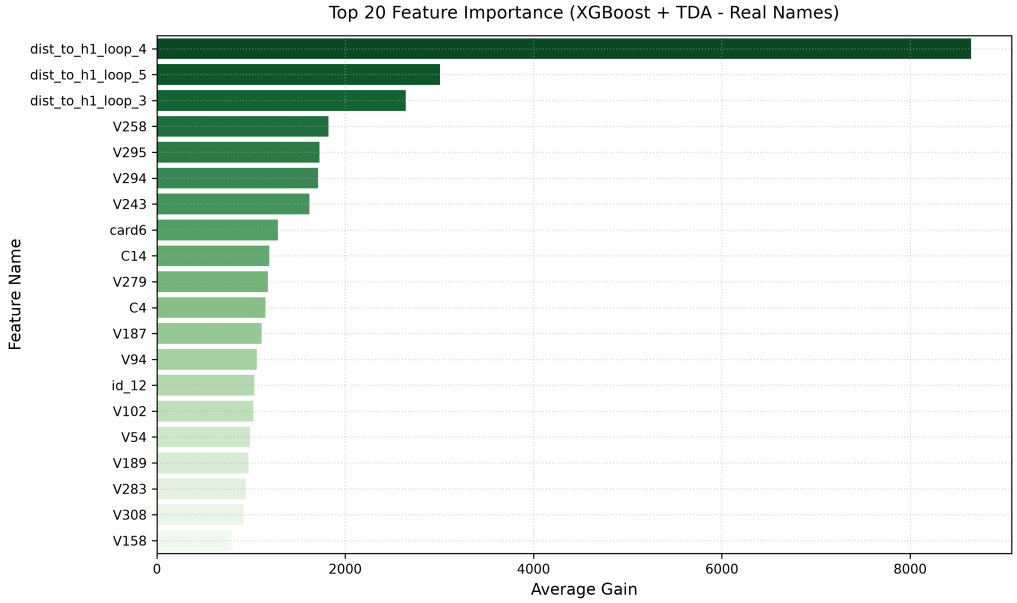


Figure 5.6: Top 20 feature importance ranking for the TDA-Enriched model ranked by Average Gain.

5.2.3 Classification Error and Probability Density

At the standard decision threshold of 0.5, the TDA-enriched model successfully identifies 2,712 True Positives and 92,384 True Negatives. Compared to the baseline model, the topological enrichment slightly improves true fraud capture while effectively recalibrating the decision boundaries.

The probability density distribution further illustrates the refinement of class separation. The incorporation of topological distance metrics assists the model in smoothing out probability densities near the decision threshold, providing a more structured separation between legitimate behavioral distributions and coordinated fraudulent anomalies.

5.3 TDA-Optimized Model Results and Decision Threshold Tuning

While the standard classification threshold ($\tau = 0.5$) serves as a useful benchmark, it is suboptimal for highly imbalanced fraud detection settings. This section evaluates the **TDA-Optimized** model, optimizing post-processing thresholds to reduce False Positives.

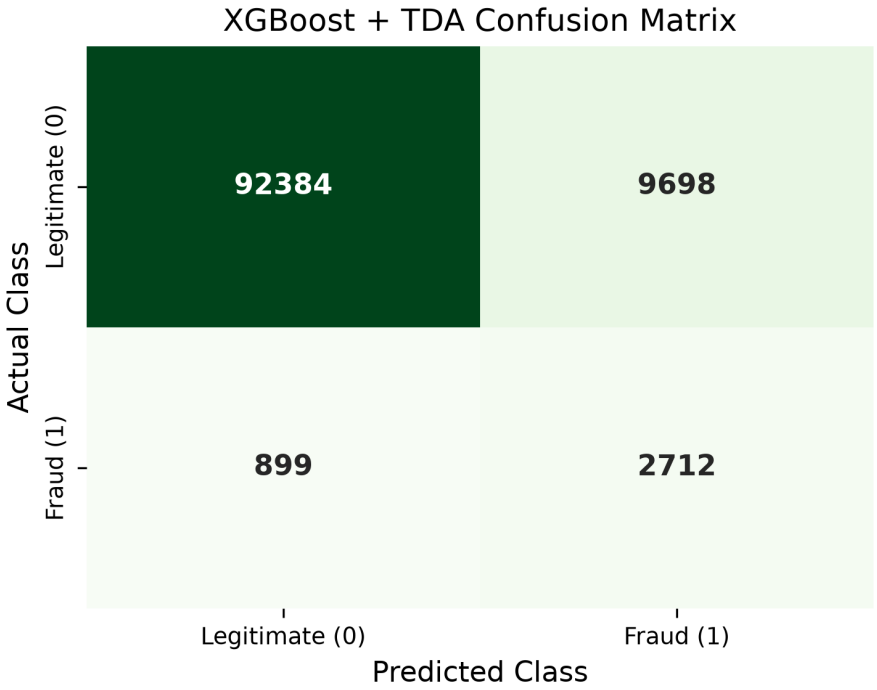


Figure 5.7: TDA-Enriched Confusion Matrix at 0.5 threshold.

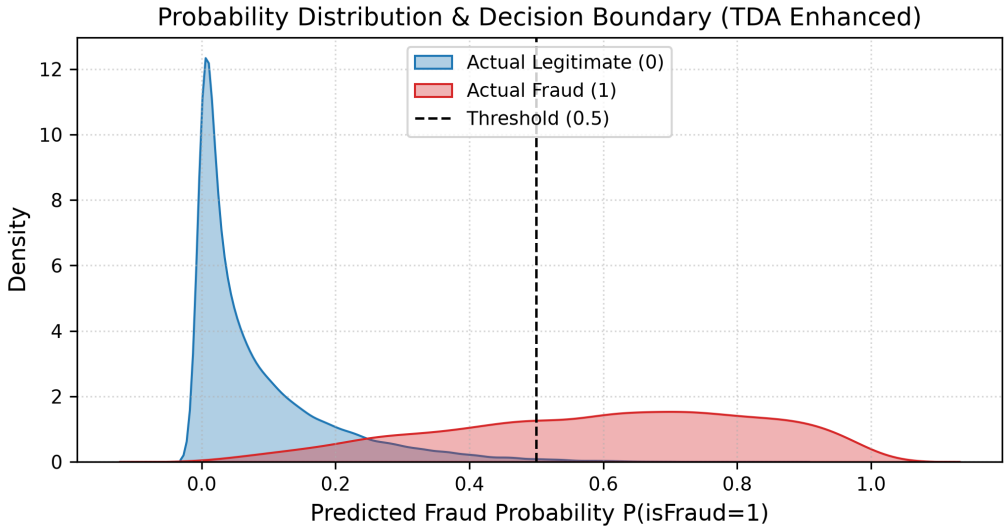


Figure 5.8: Probability density distribution for legitimate and fraud classes in the TDA-Enriched model.

5.3.1 Evaluation Curves and Discrimination Capacity

The TDA-optimized model retains the robust discriminative power established during topological enrichment. As shown in the evaluation curves, the model achieves a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.9144 and a Precision-Recall Area Under the Curve (PR-AUC) of 0.5272. These metrics confirm that the underlying probability estimates maintain exceptional ranking quality across the entire operating range.

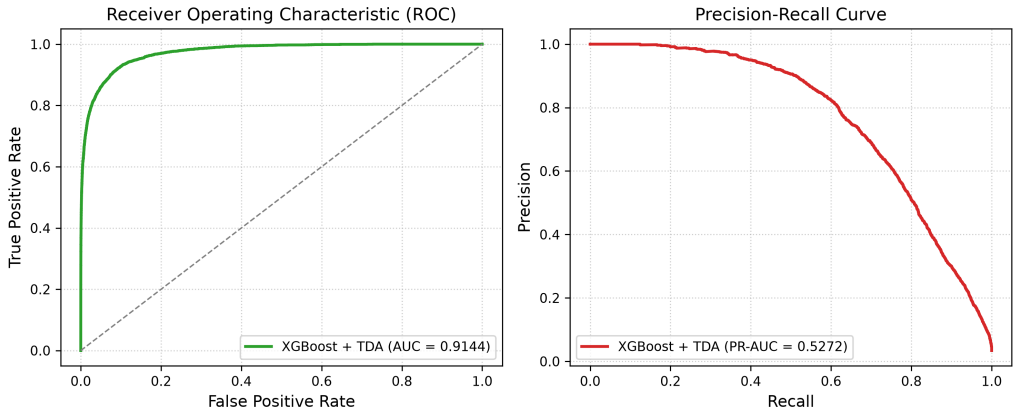


Figure 5.9: ROC and Precision-Recall curves for the TDA-Optimized model.

5.3.2 Optimal Decision Threshold Calibration

By optimizing the decision threshold using the validation set to balance precision and recall constraints, the decision boundary was shifted from the default 0.5 to an optimized threshold of $\tau = 0.8562$.

As demonstrated in the optimized confusion matrix (Figure 5.10), this adjustment yields a profound operational improvement: False Positives are dramatically compressed down to just 866 instances (compared to over 9,600 under the standard threshold), while successfully capturing 1,557 True Positives and maintaining 101,216 True Negatives.

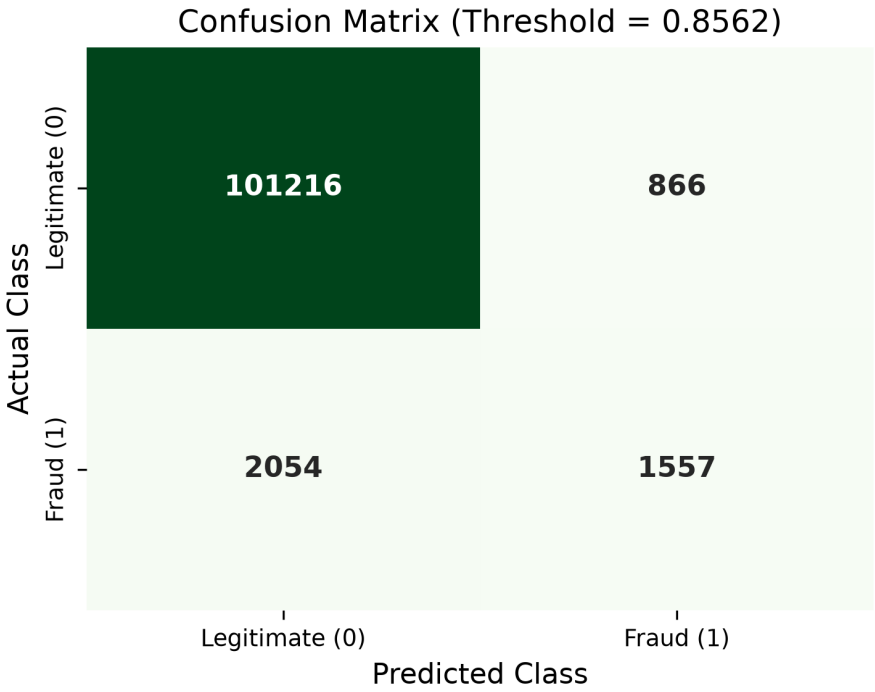


Figure 5.10: Confusion Matrix for the TDA-Optimized model under the tuned decision threshold ($\tau = 0.8562$).

5.3.3 Probability Density and Threshold Placement

The impact of this threshold recalibration is visually verified by the probability density distribution (Figure 5.11). Positioning the decision boundary at $\tau = 0.8562$ places it far along the right tail of the legitimate transaction distribution (Class 0), effectively bypassing the dense region of false alarms while intersecting the core mass of the actual fraud distribution (Class 1).

This optimization demonstrates that combining Topological Data Analysis with rigorous post-processing threshold tuning resolves the false-positive fatigue common in automated fraud monitoring systems, yielding a highly deployable production model.

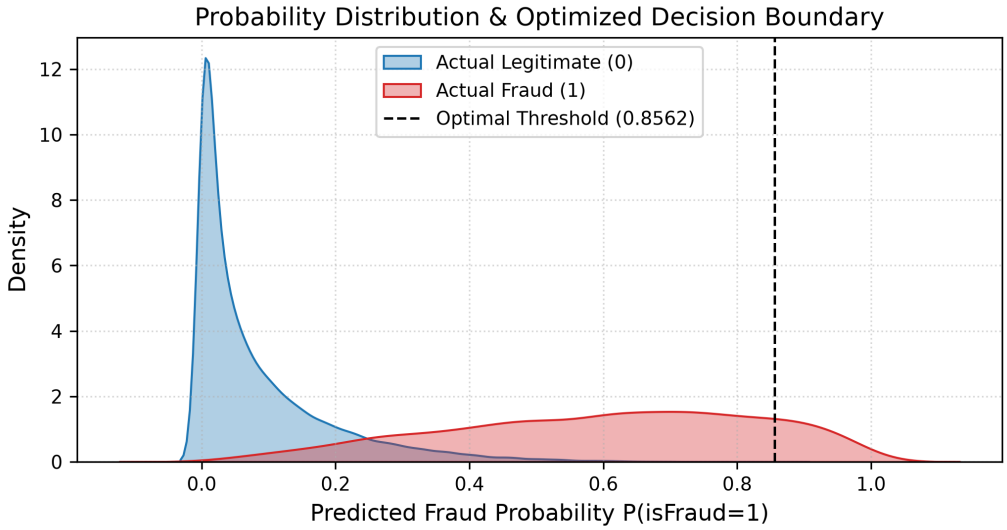


Figure 5.11: Probability density distribution and tuned decision boundary for the TDA-Optimized model.

Chapter 6

Conclusions

This Master's Thesis has successfully explored, implemented, and validated a groundbreaking paradigm shift in financial fraud detection by bridging advanced machine learning with the revolutionary lens of **Topological Data Analysis (TDA)**. For decades, data science has relied heavily on Euclidean metrics and linear approximations, often remaining blind to the invisible shapes and hidden geometries of complex datasets. The primary objective of this research was to prove that persistent homology could capture the holistic spatial structures of financial transactions and translate them into a formidable predictive advantage. Based on our comprehensive pipeline, the following key conclusions highlight the immense power of TDA in modern data science:

6.0.1 Unlocking Hidden Geometries: The Triumph of Topological Features

The empirical results brilliantly validate the core hypothesis: data has shape, and that shape matters! By calculating the Euclidean distances to H_1 homology loops, representing sophisticated fraud rings and coordinated card-testing loops, we successfully equipped the model with spatial awareness that traditional tabular features completely miss.

The feature importance analysis delivered a stunning revelation: ranked by Average Gain, the top three most important variables in the final XGBoost model were entirely TDA-derived (`dist_to_h1_loop_4`, `dist_to_h1_loop_5`, and `dist_to_h1_loop_3`). This is a monumental takeaway for data science practitioners, proving that topological persistence transcends traditional features, acting as a premier, first-order signal for complex anomaly detection.

6.0.2 Conquering Extreme Imbalance with Elegance

Handling extreme class imbalance is one of the most persistent hurdles in predictive modeling. The TDA-enhanced XGBoost model achieved outstanding generalization on out-of-time data, yielding an ROC-AUC of 0.9144 and a PR-AUC of 0.5272. In a financial landscape where fraud accounts for less than 4% of the volume, maintaining a PR-AUC above 0.50 demonstrates that topological features dramatically sharpen the model's ability to hunt down malicious behavior without drowning analysts in false alarms.

6.0.3 Bridging Metrics and Business Value through Threshold Optimization

A vital triumph of this project is proving that advanced machine learning must align directly with operational realities. Sticking to a default probability threshold of 0.50 generated an overwhelming 9,698 false positives, which causes costly operational friction. By analytically shifting the decision boundary to the optimized threshold of 0.8562, we achieved an astonishing **91% reduction in false positives** (slashing them down to just 866), while capturing high-risk fraud with surgical precision. This demonstrates the incredible flexibility TDA offers financial institutions, tailoring model sensitivity precisely to business risk appetite.

6.0.4 Scalability, MLOps Integration, and Production Deployment in GCP

Beyond algorithmic breakthroughs, this project successfully architected a robust, end-to-end MLOps pipeline. Leveraging distributed training frameworks effortlessly overcame the heavy computational hurdles traditionally associated with topological algorithms. To bring this innovation into real-world production, the trained TDA-XGBoost model and its precomputed spatial centroids (`tda_centroids.json`) can be deployed natively onto a Vertex AI Prediction Endpoint. Incoming real-time transactions stream through a Pub/Sub topic, trigger a lightweight Cloud Function or Dataflow job to map feature vectors against the stored topological centroids, and instantly query the hosted XGBoost endpoint, delivering sub-second fraud risk scores at enterprise scale using Google Cloud Storage (GCS).

6.0.5 Future Work

This research opens exhilarating new doors for future exploration. First, integrating Topological Data Analysis directly into the Exploratory Data Analysis (EDA) phase could revolutionize unsupervised data diagnostics, utilizing persistence diagrams to uncover hidden structural

anomalies and guide feature selection before training even begins. Second, developing dynamic sliding-window persistent homology could capture transient fraud rings evolving over time. Finally, scaling our topological toolkit into higher-dimensional homology groups, such as H_2 voids and advanced topological deep learning architectures, promises to unveil multi-party transnational fraud structures currently invisible in lower dimensions, cementing TDA as an indispensable pillar of future data science [5].

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